Experiment 4: Statistical Hypothesis Testing Using Python

Mapped CO(s): CO2

Aim:

To perform statistical hypothesis testing on the Titanic dataset using Python and interpret the results using p-values.

Objectives:

1. To understand the concept of hypothesis testing.
2. To formulate null and alternative hypotheses.
3. To perform statistical tests using Python.
4. To interpret p-values and significance levels.
5. To draw conclusions based on the test results.

Prerequisites:

- Basic knowledge of Python programming.
- Basic understanding of probability and statistics.
- Python 3.x installed.
- Jupyter Notebook or Google Colab.
- Required libraries: Pandas, NumPy, SciPy, Matplotlib, and Seaborn.

Theory/Background

Statistical Hypothesis Testing is a method used to determine whether there is sufficient evidence to support a claim about a population based on sample data.

A hypothesis test consists of two hypotheses:

- **Null Hypothesis (H_0):** Assumes there is no significant difference or relationship between variables.
- **Alternative Hypothesis (H_1):** Assumes there is a significant difference or relationship between variables.

The **p-value** is compared with the significance level ($\alpha = 0.05$):

- If **p-value < 0.05**, reject the null hypothesis (statistically significant difference).
- If **p-value $\geq$ 0.05**, fail to reject the null hypothesis (no statistically significant difference).

In this experiment, an **Independent Sample t-test** is performed to determine whether the average age of passengers differs significantly based on survival status.

Step 1: Import Required Python Libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

# Set style for visualizations
sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (8, 5)
```

Step 2: Load the Titanic Dataset

In [2]:
```
# Load the Titanic dataset from Seaborn
df = sns.load_dataset('titanic')
print(f"Original dataset shape: {df.shape}")
df.head()
```

```
Original dataset shape: (891, 15)
```

Out[2]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adul
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

Step 3: Select Required Variables and Clean Missing Values

We focus on the `age` and `survived` attributes. We check for missing values in the `age` column and drop any rows containing missing values.

In [3]:
```
# Check for missing values in age and survived
print("Missing values:")
print(df[['age', 'survived']].isnull().sum())

# Clean missing values by dropping rows where age is missing
df_clean = df[['age', 'survived']].dropna()
print(f"\nCleaned dataset shape: {df_clean.shape}")
df_clean.head()
```

```
Missing values:
age         177
survived      0
dtype: int64

Cleaned dataset shape: (714, 2)
```

Out[3]:

	age	survived
0	22.0	0
1	38.0	1
2	26.0	1
3	35.0	1
4	35.0	0

Step 4: Separate Passengers into Survived and Not Survived Groups

In [4]:

```
# Separate groups
age_survived = df_clean[df_clean['survived'] == 1]['age']
age_not_survived = df_clean[df_clean['survived'] == 0]['age']

# Calculate and display descriptive statistics
print(f"Number of survivors with age data: {len(age_survived)}")
print(f"Mean age of survivors: {age_survived.mean():.4f} years")
print(f"Standard deviation of survivors' age: {age_survived.std():.4f} years\n")

print(f"Number of non-survivors with age data: {len(age_not_survived)}")
print(f"Mean age of non-survivors: {age_not_survived.mean():.4f} years")
print(f"Standard deviation of non-survivors' age: {age_not_survived.std():.4f} y
```

```
Number of survivors with age data: 290
Mean age of survivors: 28.3437 years
Standard deviation of survivors' age: 14.9510 years

Number of non-survivors with age data: 424
Mean age of non-survivors: 30.6262 years
Standard deviation of non-survivors' age: 14.1721 years
```

Step 5: Perform the Independent Sample t-test using SciPy

We formulate the hypotheses:

- **Null Hypothesis (H_0):** There is no significant difference in the average age of passengers who survived and those who did not survive ($\mu_{survived} = \mu_{not\_survived}$).
- **Alternative Hypothesis (H_1):** There is a significant difference in the average age of passengers who survived and those who did not survive ($\mu_{survived} \neq \mu_{not\_survived}$).

In [5]:

```
# Perform t-test assuming equal variances
t_stat, p_val = stats.ttest_ind(age_survived, age_not_survived, equal_var=True)
```

```
print(f"t-statistic: {t_stat:.4f}")
print(f"p-value: {p_val:.4f}")
```

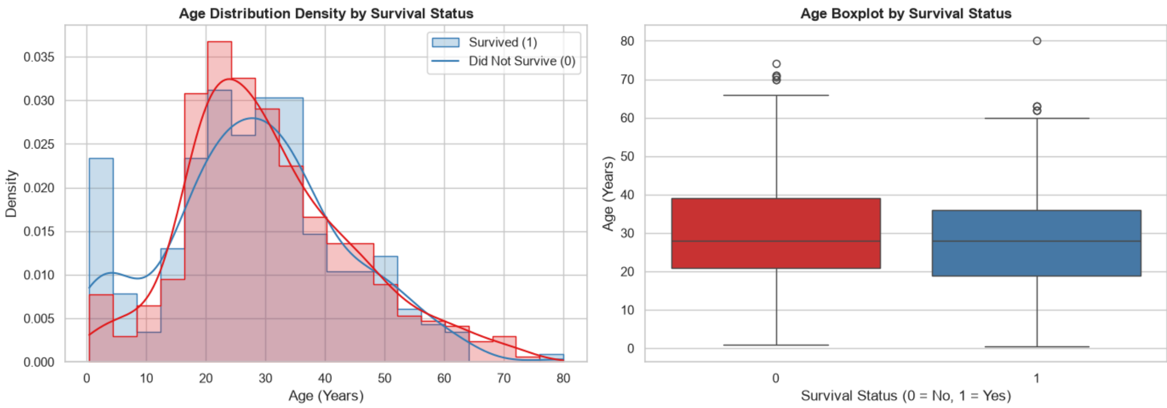
```
t-statistic: -2.0667
p-value: 0.0391
```

Step 6: Visualizing the Data

We visualize the age distributions using a combination of a boxplot and a KDE plot to understand the visual difference between the two groups.

In [6]:
```
# Plotting age distributions
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# KDE Plot
sns.histplot(data=df_clean, x='age', hue='survived', kde=True, element='step', s
axes[0].set_title('Age Distribution Density by Survival Status', fontsize=12, fo
axes[0].set_xlabel('Age (Years)')
axes[0].set_ylabel('Density')
axes[0].legend(labels=['Survived (1)', 'Did Not Survive (0)'])

# Box Plot
sns.boxplot(data=df_clean, x='survived', y='age', palette='Set1', ax=axes[1])
axes[1].set_title('Age Boxplot by Survival Status', fontsize=12, fontweight='bol
axes[1].set_xlabel('Survival Status (0 = No, 1 = Yes)')
axes[1].set_ylabel('Age (Years)')

plt.tight_layout()
plt.show()
```

```
C:\Users\Jeetu Mehra\AppData\Local\Temp\ipykernel_31948\3640563340.py:12: FutureW
arning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.boxplot(data=df_clean, x='survived', y='age', palette='Set1', ax=axes[1])
```



Step 7: Interpret the Hypothesis Test Result

In [7]:
```
alpha = 0.05
print(f"Significance Level (alpha): {alpha}")
print(f"Obtained p-value: {p_val:.4f}")
```

```python
if p_val < alpha:
    print("Conclusion: Reject the Null Hypothesis (H0). There is a statistically
else:
    print("Conclusion: Fail to Reject the Null Hypothesis (H0). There is no stat
```

```
Significance Level (alpha): 0.05
Obtained p-value: 0.0391
Conclusion: Reject the Null Hypothesis (H0). There is a statistically significant
difference in the average age of survivors versus non-survivors.
```

Exercises/Assignments

Exercise 1: Perform a t-test using the Fare variable instead of Age

- **Null Hypothesis (H_0):** There is no significant difference in the average ticket fare of passengers who survived and those who did not survive ($\mu_{survived} = \mu_{not\_survived}$).
- **Alternative Hypothesis (H_1):** There is a significant difference in the average ticket fare of passengers who survived and those who did not survive ($\mu_{survived} \neq \mu_{not\_survived}$).

In [8]:
```python
# Select fare and survived variables and drop missing values
df_fare = df[['fare', 'survived']].dropna()
print(f"Cleaned fare dataset shape: {df_fare.shape}")

# Separate groups
fare_survived = df_fare[df_fare['survived'] == 1]['fare']
fare_not_survived = df_fare[df_fare['survived'] == 0]['fare']

print(f"Mean fare of survived: ${fare_survived.mean():.4f}")
print(f"Mean fare of not survived: ${fare_not_survived.mean():.4f}\n")

# Perform independent t-test (equal_var=False because variances in fare are like
t_stat_fare, p_val_fare = stats.ttest_ind(fare_survived, fare_not_survived, equa
print(f"t-statistic (unequal variance t-test): {t_stat_fare:.4f}")
print(f"p-value: {p_val_fare:.4g}")
```

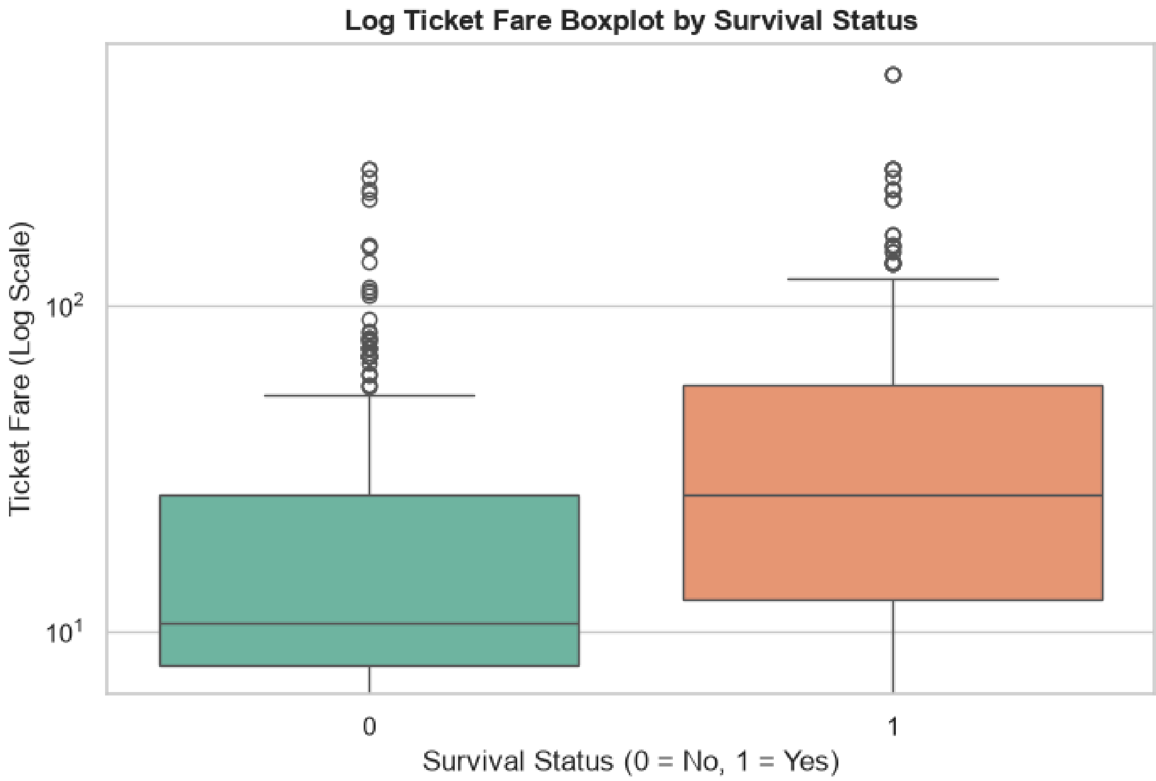
```
Cleaned fare dataset shape: (891, 2)
Mean fare of survived: $48.3954
Mean fare of not survived: $22.1179

t-statistic (unequal variance t-test): 6.8391
p-value: 2.699e-11
```

Exercise 2 & 3: Change the significance level from 0.05 to 0.01 and compare/interpret the results

In [9]:
```python
# Visualizing fare distributions
plt.figure(figsize=(8, 5))
sns.boxplot(data=df_fare, x='survived', y='fare', palette='Set2')
plt.yscale('log') # use log scale because fares are highly skewed
plt.title('Log Ticket Fare Boxplot by Survival Status', fontsize=12, fontweight=
plt.xlabel('Survival Status (0 = No, 1 = Yes)')
```

```
plt.ylabel('Ticket Fare (Log Scale)')
plt.show()

# Interpretation for Significance Level Comparison
alpha_new = 0.01
print(f"New Significance Level (alpha): {alpha_new}")
print(f"Obtained p-value for Fare: {p_val_fare:.4g}")

if p_val_fare < alpha_new:
    print("\nConclusion at alpha=0.01: Reject the Null Hypothesis (H0). The diff
else:
    print("\nConclusion at alpha=0.01: Fail to Reject the Null Hypothesis (H0)."
```

```
C:\Users\Jeetu Mehra\AppData\Local\Temp\ipykernel_31948\935004529.py:3: FutureWar
ning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.boxplot(data=df_fare, x='survived', y='fare', palette='Set2')
```




```
New Significance Level (alpha): 0.01
Obtained p-value for Fare: 2.699e-11

Conclusion at alpha=0.01: Reject the Null Hypothesis (H0). The difference in mean
ticket fare between survivors and non-survivors is highly significant.
```

Exercise 4: Apply another statistical test such as the Chi-Square test

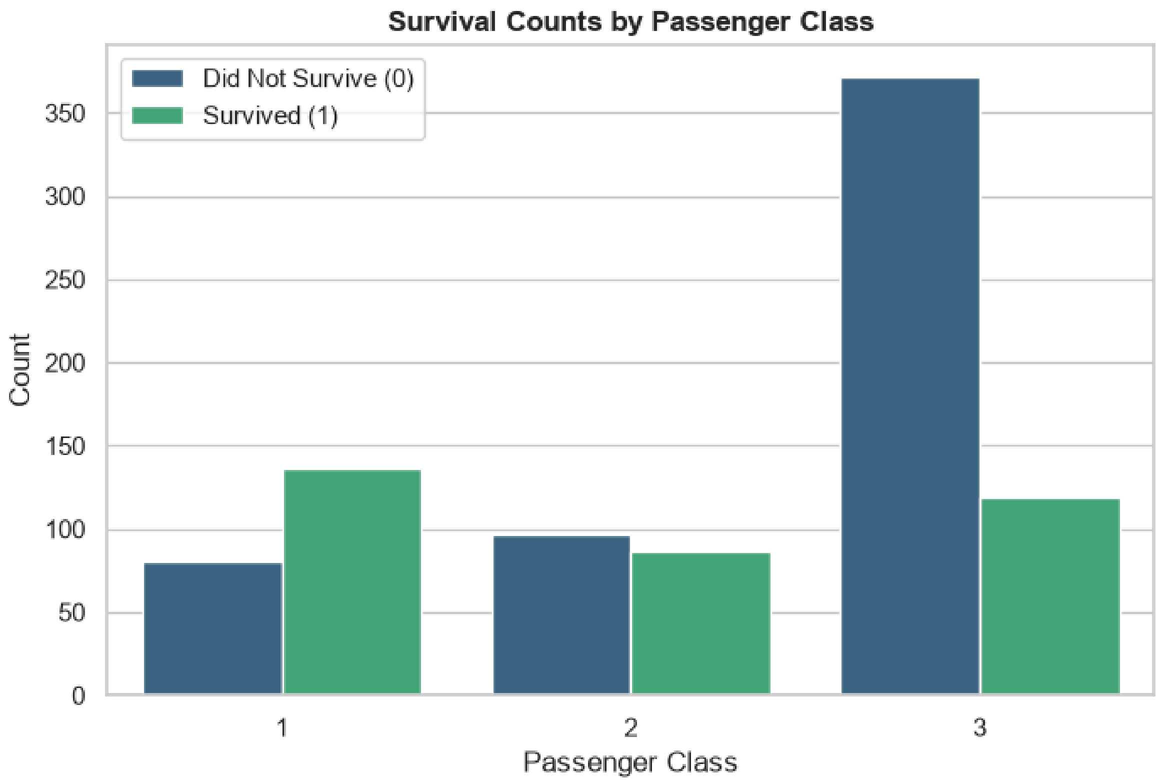
We will test whether there is a statistically significant association between survival status (`survived`) and passenger class (`pclass`).

- **Null Hypothesis (H_0):** Survival status and passenger class are independent (no association).

- **Alternative Hypothesis (H_1):** Survival status and passenger class are associated (a passenger's class affects their survival probability).

In [10]:
```
# Create a contingency table
contingency_table = pd.crosstab(df['survived'], df['pclass'])
print("Contingency Table (Observed Frequencies):")
print(contingency_table)

# Plot survival count by passenger class
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='pclass', hue='survived', palette='viridis')
plt.title('Survival Counts by Passenger Class', fontsize=12, fontweight='bold')
plt.xlabel('Passenger Class')
plt.ylabel('Count')
plt.legend(labels=['Did Not Survive (0)', 'Survived (1)'])
plt.show()

# Perform the Chi-Square Test of Independence
chi2_stat, p_val_chi2, dof, expected = stats.chi2_contingency(contingency_table)
print(f"Chi-Square Statistic: {chi2_stat:.4f}")
print(f"Degrees of Freedom: {dof}")
print(f"p-value: {p_val_chi2:.4g}\n")

print("Expected Frequencies (under the Null Hypothesis):")
print(pd.DataFrame(expected, index=['survived=0', 'survived=1'], columns=['Class

# Conclusion
alpha_chi = 0.05
if p_val_chi2 < alpha_chi:
    print(f"\nSince p-value ({p_val_chi2:.4g}) < alpha ({alpha_chi}), we reject
else:
    print(f"\nSince p-value ({p_val_chi2:.4g}) >= alpha ({alpha_chi}), we fail t
```

```
Contingency Table (Observed Frequencies):
pclass      1   2    3
survived
0          80  97  372
1         136  87  119
```




```
Chi-Square Statistic: 102.8890
Degrees of Freedom: 2
p-value: 4.549e-23

Expected Frequencies (under the Null Hypothesis):
            Class 1  Class 2  Class 3
survived=0   133.09   113.37   302.54
survived=1    82.91    70.63   188.46

Since p-value (4.549e-23) < alpha (0.05), we reject the Null Hypothesis (H0). Sur
vival status is significantly associated with passenger class.
```

Exercise 5: State the null and alternative hypotheses for a different variable

We will formulate hypotheses for **Gender/Sex** and test them using a Chi-Square test of independence.

- **Null Hypothesis (H_0):** Passenger survival status and passenger gender (`sex`) are independent (there is no significant association between gender and survival).
- **Alternative Hypothesis (H_1):** Passenger survival status and passenger gender (`sex`) are not independent (there is a significant association between gender and survival, reflecting the 'women and children first' protocol).

In [11]:

```
# Contingency table for survival vs sex
sex_contingency = pd.crosstab(df['survived'], df['sex'])
print("Contingency Table (Survival vs Sex):")
print(sex_contingency)

# Perform the Chi-Square Test
chi2_sex, p_val_sex, dof_sex, expected_sex = stats.chi2_contingency(sex_continge
print(f"\nChi-Square Statistic: {chi2_sex:.4f}")
print(f"p-value: {p_val_sex:.4g}")
```

```python
# Conclusion
if p_val_sex < 0.05:
    print("\nConclusion: Reject the Null Hypothesis (H0). There is a highly sign
else:
    print("\nConclusion: Fail to Reject the Null Hypothesis (H0).")
```

```
Contingency Table (Survival vs Sex):
sex       female  male
survived
0             81   468
1            233   109

Chi-Square Statistic: 260.7170
p-value: 1.197e-58

Conclusion: Reject the Null Hypothesis (H0). There is a highly significant associ
ation between survival status and passenger gender.
```